

SUMMARY

Applied scientist and research engineer with 6+ years of experience taking multimodal models from data design and pre/post-training through rigorous evaluation and production deployment. Led end-to-end VLM and visual-grounding projects and teams of up to eight people, built scalable PyTorch training/serving systems, and contributed 40+ pull requests to Hugging Face Transformers.

EXPERIENCE

- **Toss Bank** Seoul, South Korea
Data Scientist *Jan 2026 - Present*
 - **On-premise AI Agent System:** Building on-premise tool-calling agents that route requests between direct question answering and RAG over internal documents retrieved with image embeddings.
 - **AI Authentication – My ID MFR:** Developing My ID MFR for mobile ID card recognition and facial verification with ONNX models running through WebGPU; combining face anti-spoofing, multitask depth/normal/keypoint estimation, and face-embedding similarity.
 - **Language Model Evaluation:** Building an in-house language model benchmark aligned with public evaluation protocols and scoring heuristics to compare Qwen and GLM models and analyze performance differences.
 - **[Lead] End-to-End Document AI:** Led the full research and production pipeline for an approximately 1B-parameter vision-language model, spanning multi-page classification, orientation, layout detection, OCR/table understanding, and LLM-based key-value extraction; post-trained the model, designed stage-level diagnostics, and reached 61% exact-match accuracy for automation-ready outputs.
- **SuperbAI** Seoul, South Korea
Machine Learning Engineer *Sep 2021 - Jan 2026*
 - **[Lead] Multimodal Foundation Model:** Held approximately one year of team-leadership responsibility, leading two ML engineers through multimodal pre-training and staged text/image alignment for a visual-grounding model; curated 1.1M images, then drove evaluation and deployment.
 - **CVPR 2025 Competitions:** Won 2nd place in IOD and 4th place in FSOD challenges, validating the model on open evaluation benchmarks.
 - **[Lead] ML Training & Serving Platform:** Built reusable distributed multi-GPU training, evaluation, and deployment infrastructure with PyTorch and AWS Batch; optimized production serving with TensorRT/Triton, improving throughput by 5x and delivering 1,100+ models and 60+ customer endpoints in one year.
 - **Post-training and Efficiency:** Applied LoRA, Adapter, and LST methods and memory-efficient training, reducing GPU memory by 65.6% and training time by 44.3% while preserving model performance.
 - **Interactive Segmentation:** Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2, increasing segmentation labeling speed by 25x.
- **Kakao Enterprise** Pangyo, South Korea
Machine Learning Engineer Intern *Feb 2021 - Aug 2021*
 - **AutoML:** Fixed AutoGluon NeuralNetFastAI scaling issue and developed a Flask-based training/inference AutoML framework with a simple front-end.
- **Team ISLAND** Seoul, South Korea
Chief Technology Officer *Mar 2019 - Oct 2020*
 - **Startup:** Co-founded the company as CTO, raised approximately \$290K, and managed a five-developer core team across Android, Unity, and deep learning; across leadership roles, led up to eight people simultaneously.
 - **Mobile ML:** Built [ZZAZZ](#), a mobile video editor using detection/segmentation, 3D mapping, and tracking; deployed lightweight vision models through TensorFlow Lite.

OPEN SOURCE

- **OpenLocalAgent**: Building small tool-calling agents from scratch in pure PyTorch using open data, with pretraining, midtraining, post-training, and evaluation across ten public agent benchmarks.
- **OCR**: Building a reproducible evaluation and improvement framework for sub-1B document VLMs, covering accuracy, calibration, robustness, and LoRA adaptation.
- **Absorbed**: Developing an open-source SwiftUI roguelike with turn-based combat, branching expeditions, spirit capture, persistent progression, and English/Korean localization.
- **Hugging Face Transformers**: Contributed 40+ pull requests to a large production codebase across multimodal and vision models, independently shipping model architectures, processors, conversion scripts, distributed training fixes, tests, and documentation.
- **Hugging Face Transformers - SAM2**: Led the addition of Segment Anything 2 support to Transformers, including model integration, image/video processors, conversion flow, documentation, and tests; credited in the official [SAM2 docs](#).
- **Hugging Face Transformers - Molmo2**: Opened Molmo2 support for Transformers and published [danielcsb/Molmo2-4B](#) on the Hugging Face Hub.

PAPERS

- **ZERO: Multi-modal Prompt-based Visual Grounding**: arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices**: IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array**: UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation**: Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease**: Sejong Science High School - Dec 31, 2013

EDUCATION

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| • KAIST (Korea Advanced Institute of Science and Technology)
<i>Professional Master in Electrical Engineering; GPA: 3.55/4.3</i> | Daejeon, South Korea
<i>Feb. 2020 – Aug. 2021</i> |
| • POSTECH (Pohang University of Science and Technology)
<i>Bachelor in Electrical Engineering; GPA: 3.4/4.3</i> | Pohang, South Korea
<i>Mar. 2015 – Feb. 2020</i> |
| • UIUC (University of Illinois Urbana-Champaign)
<i>Exchange Student; GPA: 3.6/4.0</i> | Illinois, United States
<i>Jan. 2018 – Dec. 2018</i> |

PROGRAMMING SKILLS

- **Research**: Python, PyTorch, Transformers, multimodal pre/post-training, LoRA, contrastive alignment, evaluation
- **Systems**: Linux, AWS Batch, multi-GPU training, MLflow, TensorRT, Triton, ONNX, TensorFlow Lite, Git
- **Agents**: LangChain, Langfuse, Claude Code, Codex, opencode
- **Languages**: Korean (native), English (professional)